Exercises: File management 2024-2025

Exercise 1

a) Write a program that types an array of integers on the keyboard and saves this array in binary format in a file with all permission for the user write permission for the group of the file and read only for other users.

b) Write a program which loads into memory an array of integers as generated in a). The integer file does not contain the number of elements. The program should work for any number of integer data in the file.

Exercise 2

Write a program that creates a file on disk. The program asks the user for the file name to be created and the permissions to access the file, then it creates the file and displays a message whether the file was created correctly or not. Check the creation of the file with the "ls –l" command.

Exercise 3

Write a program that determines the maximum number of files a process can open at the same time under the Linux system.

Exercise 4

Write a program that simulates running the "cat file" command, this program should simply display the contents of a file (which is assumed to be of type text) on the screen. The name of the file to display must be passed as a parameter to the command line.

Exercise 5

Make a new version of the previous program that takes into account the redirection parameter of the cat command (the> and the >>, but use other symbols like +,++). For this it is suggested to use the dup2 system call to use the same display kernel from the first version of the program (a display that will be redirected to a file).

Exercise 6

Consider the following program:

1. What is the output if toto.txt and titi.txt existent?
2. What is the difference if toto.txt does not exist? And if titi.txt does not exist?

```c
#include <stdio.h>
#include <fcntl.h> /* O_RDONLY */
#include <errno.h>
int main(){
  int fd1, fd2;
  fd1 = open("toto.txt", O_RDONLY, 0);
  close(fd1);
  fd2= open("titi.txt", O_RDONLY, 0);
  printf("fd2 = %d (errno=%d)\n", fd2, errno);
  return 0;
}
```

Exercise 7

We suppose that the file toto.txt contains the six characters: "LAMBDA". What is the output of of the following programs?

Deux open

```c
#include <stdio.h>
#include <fcntl.h>
int main() {
  int fd1, fd2; char c;
  fd1=open("toto.txt", O_RDONLY, 0);
  fd2=open("toto.txt", O_RDONLY, 0);
  read(fd1, &c, 1);
  read(fd2, &c, 1);
  printf("c = %c\n", c);
  return 0;
}
```

Après dup

```c
#include <stdio.h>
#include <fcntl.h> /* O_RDONLY */
int main(){
  int fd1, fd2; char c;
  fd1=open("toto.txt", O_RDONLY, 0);
  fd2=open("toto.txt", O_RDONLY, 0);
  read(fd2, &c, 1);
  dup2(fd2, fd1);
  read(fd1, &c, 1);
  printf("c = %c\n", c);
  return 0;
}
```

Exercise 8

Write a program equivalent to the following shell command *% ls | wc*